External Plural Effect

This notebook investigates the external plural effect (ExtPE) in German compounding as defined by Schäfer & Pankratz 2018:

> A plural on the entire compound (formally on the head constituent) might trigger the use of a pluralic linking element. We call this the external plural effect.

Background

Schäfer & Pankratz 2018 assembled a German compound dataset of N=9415 compounds built from 48 first constituent (N1s), up to ten N1 for each of the seven pluralic linkers -*n*-, -*en*-, -*e*-, -"*e*-, -"-, -*er*-, -"*er*-. All N1s are countable nouns which are productively used in compounds both with and without the pluralic linking element. For each N1, up to 100 compound **tokens** with a non-pluralic (mostly zero or -*s*-) and up to 100 compound **tokens** with the pluralic linker were retrieved from DECOW16A. Since tokens and not types were counted, there is a number of duplicated constituent combinations (CC) in the data, though sometimes with alternating linkers. Besides, the original wordforms were preserved, and so compounds both in their Sg and Pl form appear, sometimes within the duplicated CC. Thus for example, the CC 'Apfel' + 'Baum' features occurrences 'Apfelbaum', 'Apfelbäume', 'Äpfelbaum', and 'Äpfelbäume' in the data. Note that no frequency threshold on the compounds was set while sampling the data.

What Schäfer & Pankratz 2018 were interested in regarding the ExtPE was whether the formal Pl form of a compound increases the probability of the pluralic linker element to occur. Going back to the example above, the question was: if the Sg form of 'Apfel' + 'Baum' is 'Apfelbaum', will the Pl form statistically tend to prefer 'Äpfelbäume' over 'Apfelbäume'?

To investigate this effect, Schäfer & Pankratz 2018 utilized the following methodology.

1. First, they created a 2x2 contingency table over the data with the following axes: non-pluralic vs. pluralic linker, and Sg vs. Pl compound form. The resulting table is below:

	Pl	Sg
non-pluralic linker	1574	3133
pluralic linker	1474	3234

2. Second, they applied the Chi2 test to identify whether there is a relation between the two variables (the "plurality" of the linker vs. the formal number of the compound). The test produced a significant result at sig = 0.05 $(\chi2 = 4.786, df = 1, p = 0.029)$.
3. Then, they measured the effect size with Cramer's V and revealed that it is quite low at $v = 0.023$.
4. Finally, the inspection of the symmetrical (standardised) residuals revealed that the diverging data points followed the wrong direction, that is, there were more non-pluralic linkers in Pl compounds that expected by chance (-2.210).

Schäfer & Pankratz 2018 conclude: even though the p-value is (still) significant, the Cramer's V and the residual values reveal that even if there is some weak association, it is opposite from expected: compounds more often have non-pluralic linkers in the Sg form.

As a confirmation, the preceding Pankratz 2017 thesis was based on the same methodology and gave similar results for the ExtPE on a different dataset. Pankratz found only four individual N1s that undergo the ExtPE in the desired direction, and four that went in the opposite direction. The ExtPE hypothesis was still refuted because no effect was detected for the entire dataset.

Objective

When scrutinizing the dataset from Schäfer & Pankratz 2018, a few groups of compounds could be detected that could erode the overall effect over the entire dataset:

- Compounds that only appear in one number. Under no observed number variation, we cannot make judgments about the behavior of these compounds in the opposite number.
- Compounds that (potentially) undergo the internal plural effect (IntPE). Since the IntPE works on a lexical level, it should be present in both numbers, so that both in Sg and Pl forms the pluralic linker is preferred (examples from the dataset: 'Äpfelkiste' vs. 'Äpfelkisten', 'Liederabend' vs. 'Liederabende'). Thus, the IntPE may interfere with the external plural effect and make the latter undetectable.
- Compounds whose N1s may take on the pluralic linker due to other constraints (examples from the dataset: 'Katzenfell'

vs. 'Katzenfelle' because of schwa, 'Kinderbuch' vs. 'Kinderbücher' as person designation). Again, if the given N1 mostly takes a pluralic linker in its Sg form, it will most likely do so in Pl, making the ExtPE undetectable.

Hypothesis 1 was the following: *after removing these "problematic" groups from the data, the ExtPE should be observable in the remaining compounds*.

Hypothesis 2 focuses on the effect of the "problematic" groups on ExtPE: *gradually letting the problematic groups back in should weaken the ExtPE down to its vanishing*.

On the other hand, the ExtPE may not be present the standard German, since variation of type 'Apfelbaum' -- 'Äpfelbäume' is ungrammatical in standard German. If the ExtPE exists, it should be an immediate product of colloquial language, maybe even unintentional. This is indirectly supported by the fact that the compounds in the data did not undergo any frequency filtering. Therefore, we proposed the **Hypothesis 3**: *most (if not all) manifestations of the ExtPE will vanish after frequency filtering. A more severe filtering will lead to a more pronounced absence of the effect*.

Data

Convention: *(N=x)* refers to the number of compound **tokens** in the dataset, *(UniqN=x)* refers to the number of unique compound **lemmata** in the dataset, *(UniqCN=x)* refers to the number of unique constituent combinations in the dataset (regardless of the linker), *(UniqN1N=x)* refers to the number of unique N1 **lemmata** in the dataset, "compounds" refers to "compound tokens" unless specified otherwise.

We prepared three modifications of the original Schäfer & Pankratz 2018 dataset resulting after removing the problematic groups gradually.

- `ic_alt` (**I**dentical **C**onstituent combinations, **ALT**ernating): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms | $(N=2318, UniqN=1246, UniqCN=869)$.
- `ic_alt_nipe` (**I**dentical **C**onstituent combinations, **ALT**ernating, **N**o **I**nt**PE**): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms and not undergoing the IntPE | $(N=1931, UniqN=1043, UniqCN=724)$.
- `ic_alt_nipe_npll` (**I**dentical **C**onstituent combinations, **ALT**ernating, **N**o **I**nt**PE**, **N**o **PL**uralic **L**inker allowed): compounds built from non-unique constituent combinations (regardless of the linker) appearing both in the Sg and the Pl forms and not undergoing the IntPE whose N1 may not attach a pluralic linker | $(N=1236, UniqN=665, UniqCN=468, UniqN1N=30)$.

In parallel, we defined three levels of frequency (the word counts (WCs) were obtained from DeReKo following the procedure outlined in Shmalts 2026, accessed mid June 2026):

- `lfreq` (low frequency): word count (WC) > 0, as in the original data.
- `mfreq` (moderate frequency): WC $>= 20$.
- `hfreq` (high frequency): WC $>= 100$.

During the studies, each of the three datasets was dynamically filtered to conform to the three frequency levels. Thus, nine unique data samples appear in the studies.

Workflow

To evaluate the proposed hypothesis, we run four studies.

Study 1

Study 1 replicates the original ExtPE study from Schäfer & Pankratz 2018 on the nine possible data samples (three datasets x three frequency levels). It answers the general question whether the ExtPE is present (in any of the samples). Comparing the individual results from the samples will additionally give an overall idea of the impact of the problematic groups and the frequency level on ExtPE's presence and strengths.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Study 2

Study 2 expands study 1 and investigates the ExtPE for each individual N1 in each of the nine data samples, following Pankratz 2017. It aims at answering whether certain N1s are more sensitive to the ExtPE than the others.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Study 3

Study 3 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Sg form only. It contrasts the distribution in the "cleanest" `ic_alt_nipe_npll` vs the two other datasets with an increasing level of "contamination" by the problematic groups.

Study 3 contributes to **Hypothesis 2** and provides a more detailed account of the effect of the problematic groups on ExtPE. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Study 4

Study 4 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Pl form only. It contrasts the distribution in the low frequency setting vs the moderate/high frequency setting, within each of the three dataset.

Study 4 contributes to **Hypothesis 3** and provides a more detailed account of the effect of the frequency setting on ExtPE. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Results

Summary

The findings from the four studies **refute Hypothesis 1** that the ExtPE generally exists in less "contaminated" data. While providing mixed evidence both **for and against Hypothesis 2**, we decide to **refute it** because the corresponding effect is not robust: while the overall effect is significant, two out of three significant N1s belonged to a problematic group, and their removal sank the ExtPE on the corresponding dataset drastically. Conversely, the findings confirm **Hypothesis 3** that any manifestations of the ExtPE are a product of the low frequency setting (better detectable in less "contaminated data").

Note that the Cramer's V value did not cross the value of *0.1* in any of the tests. That means that, even if an effect is statistically significant, the effect size is very small.

Study 1

> Summary: Study 1 provides direct evidence **against Hypothesis 1**, provides indirect evidence **against Hypothesis 2** and provides indirect evidence **for Hypothesis 3**.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

Actual outcome: The ExtPE does not reach a statistical significance in any of the nine data samples. The p-values and the Cramer's V values are comparable across the three datasets along all three frequency settings.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Actual outcome: there is a noticeable difference between the p-values across the `ic_alt_nipe_npll` and the other two datasets (along the same frequency levels), the former being much larger, which contradicts **Hypothesis 2**. However, the Cramer's V values decrease consistently within each dataset, as the frequency level of the dataset is increased. This points out that ExtPE is much weaker in moderate/high frequency settings, which indirectly points towards the expected frequency effect on the ExtPE.

Study 2

> Summary: Study 2 provides partial direct evidence **for Hypothesis 1**, provides indirect evidence **against Hypothesis 2**, and provides indirect evidence **for Hypothesis 3**.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

Actual outcome: N1s 'Apfel' undergoes the ExtPE in the desired direction at marginal significance level sig = 0.1 (increased p-

value threshold to account for the small sample sizes), but only in `ic_alt_nipe_npll` and `ic_alt_nipe` and only in the low frequency setting ($χ2 = 3.242, df = 1, p = 0.072, v = 0.265$). N1 'Bucht' undergoes the ExtPE in the desired direction at significance level sig = 0.1 in `ic_alt_nipe` and only in the low frequency setting ($χ2 = 2.716, df = 1, p = 0.099, $v = 0.243$). N1 'Person' undergoes the ExtPE in the desired direction at significance level close to sig = 0.05 in `ic_alt_nipe` and `ic_alt`, but, again, only in the low frequency setting ($χ2 = 3.967, df = 1, p = 0.046, v = 0.245$).

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Actual outcome: the fact that the marginal significance disappears with 'Apfel' and 'Bucht' in the most "contaminated" dataset `ic_alt` indirectly points towards the effect of the problematic groups on the ExtPE. However, removal of N1s 'Person' and 'Bucht' from `ic_alt_nipe_npll` speaks strongly against the problematic groups effect, since these are two of out three N1s that contribute to the ExtPE the most. The fact that the (marginal) significance disappears with all three N1s in the moderate/high frequency settings indirectly points towards the effect of frequency on the ExtPE.

The significant results of N1s 'Apfel' and 'Person' confirm finding of Pankratz 2017. However, none of the other six "significant" N1s reported by Pankratz 2017 reach statistical significance in any of the nine data samples.

Study 3

> Summary: Study 3 provides direct evidence **for Hypothesis 2**.

Study 3 contributes to **Hypothesis 2**. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Actual outcome: while comparison of the linker distributions over the compounds in the Sg form in `ic_alt_nipe_npll` vs `ic_alt_nipe` did not produce statistical significance, comparing the "cleanest" `ic_alt_nipe_npll` and the most "contaminated" `ic_alt` reveals a statistically significant effect at sig = 0.005 ($χ2 = 12.539, df = 1, p \approx 0.0, v = 0.085$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Sg form in `ic_alt` have a larger share of pluralic linkers, and those in `ic_alt_nipe_npll` a smaller share of pluralic linkers, than expected by chance (3.593). This means that compounds in `ic_alt` have more sources for the pluralic linker to appear in the Sg form. The finding contradicts the indirect evidence against Hypothesis 2 from Studies 1 and 2 and does not confirm the effect.

Study 4

> Summary: Study 4 provides direct evidence **for Hypothesis 3** and indirect evidence **for Hypothesis 2**.

Study 4 contributes to **Hypothesis 4**. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Actual outcome: comparison of the linker distributions over the compounds in the Pl form in difference frequency settings produced statistical significance only in the `ic_alt_nipe_npll` dataset: low vs. moderate frequency reaches significance at sig = 0.05, and low vs. high frequency at sig = 0.005 (low vs. moderate: $χ2 = 4.28, df = 1, p = 0.039, v = 0.068$; low vs. high: $χ2 = 8.19, df = 1, p = 0.004, v = 0.097$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Pl form inserted pluralic linkers more frequently in the low frequency setting and less frequently in the moderate/high than expected by chance (2.138 and 2.935, respectively). This finding supports the indirect evidence for Hypothesis 3 from Studies 1 and 2, and confirms the effect.

The fact that the more "contaminated" datasets did not exhibit a significant difference additionally speaks for Hypothesis 2 and may be explained by that the additional sources of pluralic linkers there erode the effect. This is attested by the fact that, while insignificant, the p-values still increase, and the Cramer's V values decrease there consistently with increasing frequency level.

Conclusion

While the theoritized ExtPE is not strong or consistent enough to be acknowledged as an established phenomenon in German, there exist arbitrary and scattered weak manifestations thereof. These manifestations occur more frequently in the low frequency setting than in the moderate/high frequency setting, with a statistically significant difference, and are not bound to any N1s or groups of those.

Study 1

Study 1 replicates the original ExtPE study from Schäfer & Pankratz 2018 on the nine possible data samples (three datasets x three frequency levels). It answers the general question whether the ExtPE is present (in any of the samples). Comparing the individual results from the samples will additionally give an overall idea of the impact of the problematic groups and the frequency level on ExtPE's presence and strengths.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

The following table provides an overview over the results of Study 1. For each combination of the dataset and the frequency setting, the results of the Chi2 test over the resulting sample is given. We can see that none of the p-values are even near low.

```
# 3D res DF: dataset x freq x chi2 test metrics
chi2_summary = pd.concat(chi2_summaries, keys=[p[0] for p in datasets])  \
    .swaplevel(0, 1)    \
    .sort_index(level=0, sort_remaining=False)  \
    .reindex(chi2df.index, level=0)
savetsv(chi2_summary, st1dir, "chi2_summary")
chi2_summary
```

		lfreq	mfreq	hfreq
N	ic_alt_nipe_npll	1074.000	773.000	666.000
	ic_alt_nipe	1930.000	1248.000	1076.000
	ic_alt	2317.000	1516.000	1308.000
df	ic_alt_nipe_npll	1.000	1.000	1.000
	ic_alt_nipe	1.000	1.000	1.000
	ic_alt	1.000	1.000	1.000
chi2	ic_alt_nipe_npll	0.697	0.002	0.162
	ic_alt_nipe	2.051	0.139	0.000
	ic_alt	1.884	0.175	0.012
p	ic_alt_nipe_npll	0.404	0.962	0.687
	ic_alt_nipe	0.152	0.710	1.000
	ic_alt	0.170	0.676	0.914
cramer_v	ic_alt_nipe_npll	0.025	0.002	0.016
	ic_alt_nipe	0.033	0.011	0.000
	ic_alt	0.029	0.011	0.003
resids_cong	ic_alt_nipe_npll	0.898	0.124	-0.487
	ic_alt_nipe	1.478	0.430	0.028
	ic_alt	1.414	0.470	0.163
resids_incong	ic_alt_nipe_npll	-0.898	-0.124	0.487
	ic_alt_nipe	-1.478	-0.430	-0.028
	ic_alt	-1.414	-0.470	-0.163

We can also observe that there is a noticeable difference between the p-values across the `ic_alt_nipe_npll` and the other two datasets (along the same frequency levels), the former being much larger. As wil be explained in Study, this is because two of the three significant N1s are not present in `ic_alt_nipe_npll`. However, there is a clear trend toward increasing p-value with the increasing frequency level, for all three datasets. Similarly, the Cramer's V values drop consistently as the frequency level is increased.

```
1 chi2_summary.loc[["p", "cramer_v"]]
```

		lfreq	mfreq	hfreq
p	ic_alt_nipe_npll	0.404	0.962	0.687
	ic_alt_nipe	0.152	0.710	1.000
	ic_alt	0.170	0.676	0.914
cramer_v	ic_alt_nipe_npll	0.025	0.002	0.016
	ic_alt_nipe	0.033	0.011	0.000
	ic_alt	0.029	0.011	0.003

Study 1 Results

> Summary: Study 1 provides direct evidence **against Hypothesis 1**, provides indirect evidence **against Hypothesis 2** and provides indirect evidence **for Hypothesis 3**.

Study 1 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting.

Actual outcome: The ExtPE does not reach a statistical significance in any of the nine data samples. The p-values and the Cramer's V values are comparable across the three datasets along all three frequency settings.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE.

Actual outcome: there is a noticeable difference between the p-values across the `ic_alt_nipe_npll` and the other two datasets (along the same frequency levels), the former being much larger, which contradicts the problematic group effect. However, the Cramer's V values decrease consistently within each dataset, as the frequency level of the dataset is increased. This points out that ExtPE is much weaker in moderate/high frequency settings, which indirectly points towards the expected frequency effect on the ExtPE.

Study 2

Study 2 expands study 1 and investigates the ExtPE for each individual N1 in each of the nine data samples, following Pankratz 2017. It aims at answering whether certain N1s are more sensitive to the ExtPE than the others.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Only three N1s reached a marginal statistical significance at sig = 0.1 (increased p-value threshold to account for the small sample sizes).

Below is a table with the results of the Chi2 test for N1s that achieved a marginal statistical significance at sig = 0.1 (increased p-value threshold to account for the smaller sample sizes).

```
chi2_summary = pd.concat(chi2_summaries_conc, axis=1)
savetsv(chi2_summary, st2dir, "chi2_summary")
chi2_summary
```

		apfel			bucht	person
		lfreq	mfreq	hfreq	lfreq	lfreq
N	ic_alt_nipe_npll	46.000	33.000	26.000	NaN	NaN
	ic_alt_nipe	46.000	33.000	26.000	46.000	51.000
	ic_alt	52.000	33.000	26.000	49.000	66.000
df	ic_alt_nipe_npll	1.000	1.000	1.000	NaN	NaN
	ic_alt_nipe	1.000	1.000	1.000	1.000	1.000
	ic_alt	1.000	1.000	1.000	1.000	1.000
chi2	ic_alt_nipe_npll	3.242	0.136	1.187	NaN	NaN
	ic_alt_nipe	3.242	0.136	1.187	2.716	3.780
	ic_alt	2.233	0.136	1.187	2.366	3.967
p	ic_alt_nipe_npll	0.072	0.712	0.276	NaN	NaN
	ic_alt_nipe	0.072	0.712	0.276	0.099	0.052
	ic_alt	0.135	0.712	0.276	0.124	0.046
cramer_v	ic_alt_nipe_npll	0.265	0.064	0.214	NaN	NaN
	ic_alt_nipe	0.265	0.064	0.214	0.243	0.272
	ic_alt	0.207	0.064	0.214	0.220	0.245
resids_cong	ic_alt_nipe_npll	2.107	0.821	1.323	NaN	NaN
	ic_alt_nipe	2.107	0.821	1.323	1.951	2.228
	ic_alt	1.776	0.821	1.323	1.836	2.246
resids_incong	ic_alt_nipe_npll	-2.107	-0.821	-1.323	NaN	NaN
	ic_alt_nipe	-2.107	-0.821	-1.323	-1.951	-2.228
	ic_alt	-1.776	-0.821	-1.323	-1.836	-2.246
p_sidak	ic_alt_nipe_npll	0.072	0.712	0.276	NaN	NaN
	ic_alt_nipe	0.072	0.712	0.276	0.099	0.052
	ic_alt	0.135	0.712	0.276	0.124	0.046

'Apfel' reaches marginal significance only in the `ic_alt_nipe_npll` and `ic_alt_nipe` datasets, and only in the low frequency setting. 'Bucht' reaches marginal significance only in the `ic_alt_nipe` dataset, and only in the low frequency setting. 'Person' reaches marginal significance only in the `ic_alt_nipe` and the `ic_alt` datasets, and only in the low frequency setting.

```
chi2_summary.loc[["p", "cramer_v"]]
```

		apfel			bucht	person
		lfreq	mfreq	hfreq	lfreq	lfreq
p	ic_alt_nipe_npll	0.072	0.712	0.276	NaN	NaN
	ic_alt_nipe	0.072	0.712	0.276	0.099	0.052
	ic_alt	0.135	0.712	0.276	0.124	0.046
cramer_v	ic_alt_nipe_npll	0.265	0.064	0.214	NaN	NaN
	ic_alt_nipe	0.265	0.064	0.214	0.243	0.272
	ic_alt	0.207	0.064	0.214	0.220	0.245

The reason N1s 'Person' and 'Bucht' have only an `lfreq` column is because after frequency filtering, the remaining compounds with these N1s ceased to alternate between non-pluralic and pluralic linker in different numbers, always inserting the latter. That led to corrupted contingency tables, so no result could be obtained for these N1s in the moderate/high frequency setting. It is, however, clear that no alternation means no ExtPE whatsoever.

Furthermore, N1s 'Person' and 'Bucht' were removed from the `ic_alt_nipe_npll` dataset as they may attach the PLL under non-pluralic constraints. Being two of the only three significant N1s found in the data, these N1s should contribute noticeably into the overall effect, which explains why the "cleanest" `ic_alt_nipe_npll` has the weakest ExtPE. This speaks strongly against **Hypothesis 2**.

Study 2 Results

> Summary: Study 2 provides partial direct evidence **for Hypothesis 1**, provides indirect evidence **against Hypothesis 2**, and provides indirect evidence **for Hypothesis 3**.

Study 2 contributes to **Hypothesis 1**. Expected outcome: ExtPE will be confirmed on less "contaminated" datasets in the lower frequency setting with some N1s.

Actual outcome: N1s 'Apfel' undergoes the ExtPE in the desired direction at marginal significance level sig = 0.1 (increased p-value threshold to account for the small sample sizes), but only in `ic_alt_nipe_npll` and `ic_alt_nipe` and only in the low frequency setting ($χ2 = 3.242, df = 1, p = 0.072, v = 0.265$$χ2 = 3.242, df = 1, p = 0.072, v = 0.265$). N1 'Bucht' undergoes the ExtPE in the desired direction at significance level sig = 0.1 in `ic_alt_nipe` and only in the low frequency setting ($χ2 = 2.716, df = 1, p = 0.099, $$χ2 = 2.716, df = 1, p = 0.099, $v = 0.243$). N1 'Person' undergoes the ExtPE in the desired direction at significance level close to sig = 0.05 in `ic_alt_nipe` and `ic_alt`, but, again, only in the low frequency setting ($$). N1 'Person' undergoes the ExtPE in the desired direction at significance level close to sig = 0.05 in `ic_alt_nipe` and `ic_alt`, but, again, only in the low frequency setting ($χ2 = 3.967, df = 1, p = 0.046, v = 0.245$).

It also indirectly contributes to **Hypothesis 2** and **Hypothesis 3**. Expected outcome: the increase in the level of "contamination" and frequency will result in fading/vanishing ExtPE with some or all N1s.

Actual outcome: the fact that the marginal significance disappears with 'Apfel' and 'Bucht' in the most "contaminated" dataset `ic_alt` indirectly points towards the effect of the problematic groups on the ExtPE. However, removal of N1s 'Person' and 'Bucht' from `ic_alt_nipe_npll` speaks strongly against the problematic groups effect, since these are two of out three N1s that contribute to the ExtPE the most. The fact that the (marginal) significance disappears with all three N1s in the moderate/high frequency settings indirectly points towards the effect of frequency on the ExtPE.

The significant results of N1s 'Apfel' and 'Person' confirm finding of Pankratz 2017. However, none of the other six "significant" N1s reported by Pankratz 2017 reach statistical significance in any of the nine data samples.

Study 3

Study 3 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Sg form only. It contrasts the distribution in the "cleanest" `ic_alt_nipe_npll` vs the two other datasets with an increasing level of "contamination" by the problematic groups.

Study 3 contributes to **Hypothesis 2** and provides a more detailed account of the effect of the problematic groups on ExtPE. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

We find that `ic_alt` has a larger share of pluralic linker in the compounds in the Sg form, reaching a statistical significance at sig = 0.005.

```
1 chi2_summary
```

	vs_ic_alt_nipe	vs_ic_alt
N	1517.000	1729.000
df	1.000	1.000
chi2	1.430	12.539
p	0.232	0.000
cramer_v	0.031	0.085
resids_cong	1.251	3.593
resids_incong	-1.251	-3.593

Study 3 Results

> Summary: Study 3 provides direct evidence **for Hypothesis 2**.

Study 3 contributes to **Hypothesis 2**. Expected outcome: more "contaminated" datasets will have a larger share of pluralic linkers in the Sg form (due to more possible sources).

Actual outcome: while comparison of the linker distributions over the compounds in the Sg form in `ic_alt_nipe_npll` vs `ic_alt_nipe` did not produce statistical significance, comparing the "cleanest" `ic_alt_nipe_npll` and the most "contaminated" `ic_alt` reveals a statistically significant effect at sig = 0.005 ($χ2 = 12.539, df = 1, p \approx 0.0, v = 0.085$$χ2 = 12.539, df = 1, p \approx 0.0, v = 0.085$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Sg form in `ic_alt` have a larger share of pluralic linkers, and those in `ic_alt_nipe_npll` a smaller share of pluralic linkers, than expected by chance ($3.593$$3.593$). This means that compounds in `ic_alt` have more sources for the pluralic linker to appear in the Sg form. The finding contradicts the indirect evidence against Hypothesis 2 from Studies 1 and 2 and does not confirm the effect.

Study 4

Study 4 focuses on the distribution of non-pluralic and pluralic linker in compounds in the Pl form only. It contrasts the distribution in the low frequency setting vs the moderate/high frequency setting, within each of the three dataset.

Study 4 contributes to **Hypothesis 3** and provides a more detailed account of the effect of the frequency setting on ExtPE. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

```
# 3D res DF: dataset x freq pairing x chi2 test metrics
chi2_summary = pd.concat(chi2_summaries, keys=[p[0] for p in datasets])  \
    .swaplevel(0, 1)    \
    .sort_index(level=0, sort_remaining=False)  \
    .reindex(chi2df.index, level=0)
savetsv(chi2_summary, st4dir, "chi2_summary")
chi2_summary
```

		vs_mfreq	vs_hfreq
N	ic_alt_nipe_npll	919.000	866.000
	ic_alt_nipe	1580.000	1494.000
	ic_alt	1876.000	1772.000
df	ic_alt_nipe_npll	1.000	1.000
	ic_alt_nipe	1.000	1.000
	ic_alt	1.000	1.000
chi2	ic_alt_nipe_npll	4.280	8.190
	ic_alt_nipe	1.718	2.933
	ic_alt	0.939	1.952
p	ic_alt_nipe_npll	0.039	0.004
	ic_alt_nipe	0.190	0.087
	ic_alt	0.332	0.162
cramer_v	ic_alt_nipe_npll	0.068	0.097
	ic_alt_nipe	0.033	0.044
	ic_alt	0.022	0.033
resids_cong	ic_alt_nipe_npll	2.138	2.935
	ic_alt_nipe	1.363	1.767
	ic_alt	1.016	1.446
resids_incong	ic_alt_nipe_npll	-2.138	-2.935
	ic_alt_nipe	-1.363	-1.767
	ic_alt	-1.016	-1.446

The only comparison that produces statistically significant results at sig = 0.05 is that in `ic_alt_nipe_npll`. On other datasets, p-values decrease, and the Cramer's V values decrease consistently with increasing frequency.

```
chi2_summary.loc[["p", "cramer_v"]]
```

		vs_mfreq	vs_hfreq
p	ic_alt_nipe_npll	0.039	0.004
	ic_alt_nipe	0.190	0.087
	ic_alt	0.332	0.162
cramer_v	ic_alt_nipe_npll	0.068	0.097
	ic_alt_nipe	0.033	0.044
	ic_alt	0.022	0.033

Study 4 Results

> Summary: Study 4 provides direct evidence **for Hypothesis 3** and indirect evidence **for Hypothesis 2**.

Study 4 contributes to **Hypothesis 4**. Expected outcome: the share of the pluralic linkers in compounds in the Pl form will be larger in in the low frequency setting (since most of the ExtPE manifestations are a product of immediate language).

Actual outcome: comparison of the linker distributions over the compounds in the Pl form in difference frequency settings produced statistical significance only in the `ic_alt_nipe_npll` dataset: low vs. moderate frequency reaches significance at sig = 0.05, and low vs. high frequency at sig = 0.005 (low vs. moderate: $χ2 = 4.28, df = 1, p = 0.039, v = 0.068$$χ2 = 4.28, df = 1, p = 0.039, v = 0.068$; low vs. high: $χ2 = 8.19, df = 1, p = 0.004, v = 0.097$$χ2 = 8.19, df = 1, p = 0.004, v = 0.097$). The examination of the symmetrical (standardised) residuals reveals that the diverging data points followed the expected direction, that is, compounds in the Pl form inserted pluralic linkers more frequently in the low frequency setting and less frequently in the moderate/high than expected by chance ($2.138$$2.138$ and $2.935$$2.935$, respectively). This finding supports the indirect evidence for Hypothesis 3 from Studies 1 and 2, and confirms the effect.

The fact that the more "contaminated" datasets did not exhibit a significant difference additionally speaks for Hypothesis 2 and may be explained by that the additional sources of pluralic linkers there erode the effect. This is attested by the fact that, while insignificant, the p-values still increase, and the Cramer's V values decrease there consistently with increasing frequency level.